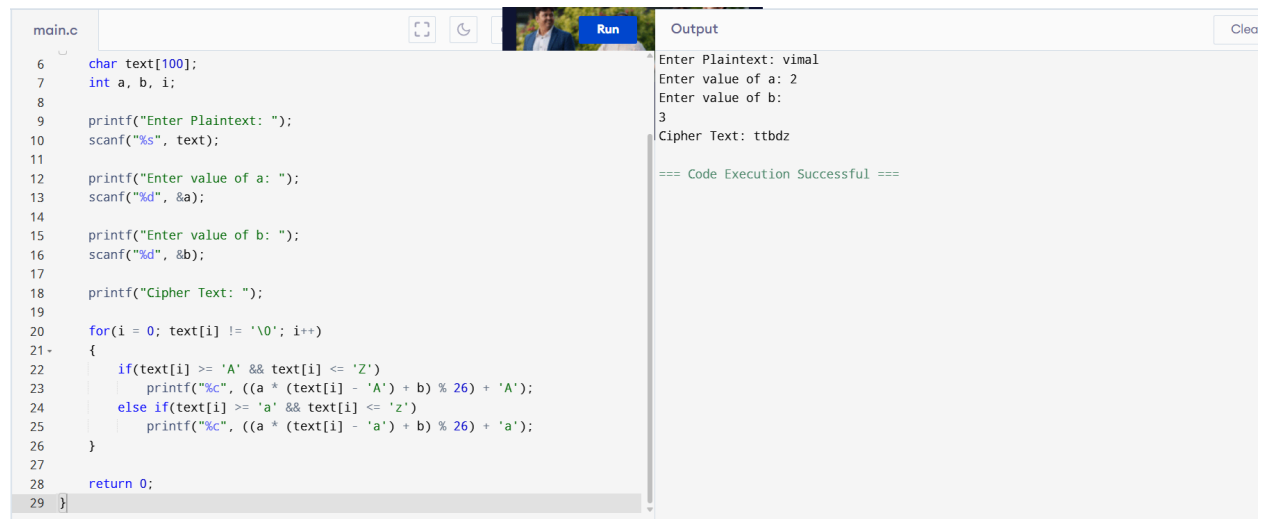


5. Affine Caesar Cipher



The screenshot shows a C program in a file named `main.c` being executed. The program prompts the user for a plaintext, a shift value `a`, and a multiplier value `b`. It then calculates the ciphertext using the Affine Caesar Cipher formula. The output shows the plaintext "vimal", `a=2`, `b=3`, and the resulting ciphertext "ttbdz".

```
main.c
6 char text[100];
7 int a, b, i;
8
9 printf("Enter Plaintext: ");
10 scanf("%s", text);
11
12 printf("Enter value of a: ");
13 scanf("%d", &a);
14
15 printf("Enter value of b: ");
16 scanf("%d", &b);
17
18 printf("Cipher Text: ");
19
20 for(i = 0; text[i] != '\0'; i++)
21 {
22     if(text[i] >= 'A' && text[i] <= 'Z')
23         printf("%c", ((a * (text[i] - 'A') + b) % 26) + 'A');
24     else if(text[i] >= 'a' && text[i] <= 'z')
25         printf("%c", ((a * (text[i] - 'a') + b) % 26) + 'a');
26 }
27
28 return 0;
29 }
```

Output

```
Enter Plaintext: vimal
Enter value of a: 2
Enter value of b: 3
Cipher Text: ttbdz

=== Code Execution Successful ===
```